Year 5 Mathematics

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Learning Strategies

Mathematics is often the most challenging subject for students. Much of the trouble comes from the fact that mathematics is about logical thinking, not memorizing rules or remembering formulas. It requires a different style of thinking than other subjects. The students who seem to be "naturally" good at math just happen to adopt the correct strategies of thinking that math requires – often they don't even realise it. We have isolated several key learning strategies used by successful maths students and have made icons to represent them. These icons are distributed throughout the book in order to remind students to adopt these necessary learning strategies:



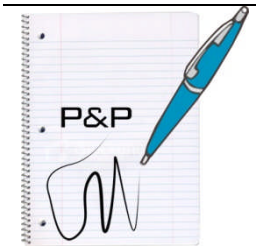
Talk Aloud Many students sit and try to do a problem in complete silence inside their heads. They think that solutions just pop into the heads of 'smart' people. You absolutely must learn to talk aloud and listen to yourself, literally to talk yourself through a problem. Successful students do this without realising. It helps to structure your thoughts while helping your tutor understand the way you think.



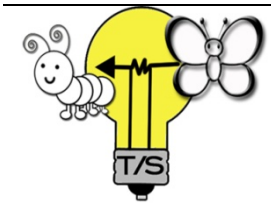
BackChecking This means that you will be doing every step of the question twice, as you work your way through the question to ensure no silly mistakes. For example with this question: $3 \times 2 - 5 \times 7$ you would do "3 times 2 is 5 ... let me check – no 3×2 is 6 ... minus 5 times 7 is minus 35 ... let me check ... minus 5×7 is minus 35. Initially, this may seem time-consuming, but once it is automatic, a great deal of time and marks will be saved.



Avoid Cosmetic Surgery Do not write over old answers since this often results in repeated mistakes or actually erasing the correct answer. When you make mistakes just put one line through the mistake rather than scribbling it out. This helps reduce silly mistakes and makes your work look cleaner and easier to backcheck.



Pen to Paper It is always wise to write things down as you work your way through a problem, in order to keep track of good ideas and to see concepts on paper instead of in your head. This makes it easier to work out the next step in the problem. Harder maths problems cannot be solved in your head alone – put your ideas on paper as soon as you have them – always!



Transfer Skills This strategy is more advanced. It is the skill of making up a simpler question and then transferring those ideas to a more complex question with which you are having difficulty.

For example if you can't remember how to do long addition because you can't recall exactly how to carry the one: $\begin{array}{r} 5889 \\ +\,4587 \\ \hline \end{array}$ then you may want to try adding numbers which you do know how to calculate that also involve carrying the one: $\begin{array}{r} 5 \\ +\,9 \\ \hline \end{array}$

This skill is particularly useful when you can't remember a basic arithmetic or algebraic rule, most of the time you should be able to work it out by creating a simpler version of the question.

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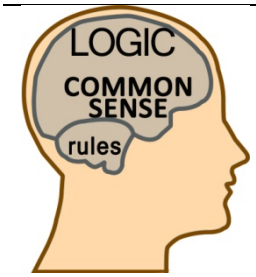
Format Skills These are the skills that keep a question together as an organized whole in terms of your working out on paper. An example of this is using the "=" sign correctly to keep a question lined up properly. In numerical calculations format skills help you to align the numbers correctly.

This skill is important because the correct working out will help you avoid careless mistakes. When your work is jumbled up all over the page it is hard for you to make sense of what belongs with what. Your "silly" mistakes would increase. Format skills also make it a lot easier for you to check over your work and to notice/correct any mistakes.

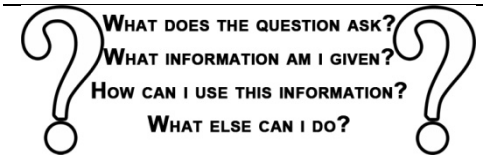
Every topic in math has a way of being written with correct formatting. You will be surprised how much smoother mathematics will be once you learn this skill. Whenever you are unsure you should always ask your tutor or teacher.



Its Ok To Be Wrong Mathematics is in many ways more of a skill than just knowledge. The main skill is problem solving and the only way this can be learned is by thinking hard and making mistakes on the way. As you gain confidence you will naturally worry less about making the mistakes and more about learning from them. Risk trying to solve problems that you are unsure of, this will improve your skill more than anything else. It's ok to be wrong – it is NOT ok to not try.



Avoid Rule Dependency Rules are secondary tools; common sense and logic are primary tools for problem solving and mathematics in general. Ultimately you must understand Why rules work the way they do. Without this you are likely to struggle with tricky problem solving and worded questions. Always rely on your logic and common sense first and on rules second, always ask Why?



Self Questioning This is what strong problem solvers do naturally when they get stuck on a problem or don't know what to do. Ask yourself these questions. They will help to jolt your thinking process; consider just one question at a time and Talk Aloud while putting Pen To Paper.

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Year 5 Mathematics

Number

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Useful formulae and hints

Roman Numerals:

V = 5

X = 10

L = 50

C = 100

Place value: In the number “abcdefg”

g represents units

f represents tens

e represents hundreds

d represents thousands

c represents tens of thousands

b represents hundreds of thousands

a represents millions

A factor is a number that divides into a given number equally. For example, the factors of 12 are 1, 2, 3, 4, 6 and 12

A multiple is a number that a given number divides into evenly. For example, the multiples of 4 are 4, 8, 12, 16, 20...

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A unit fraction shows one part out the total number of parts. For example, ½ means one part out of two

To add or subtract decimals, line up the two numbers according to their decimal points, then add or subtract as normal, carrying the decimal point down to the same place in the answer

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Exercise 1

Roman Numerals

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1) Convert the following Roman numerals to Arabic

a) V

b) X

c) C

d) D

e) L

2) Convert the following to Roman numerals

a) 10

b) 200

c) 6

d) 11

e) 105

3) Convert the following to Arabic numerals

a) LV

b) CXI

c) CLVII

d) XX

e) LXXIII

4) Convert the following to Roman numerals

a) 33

b) 56

c) 105

d) 12

e) 171

5) Convert the following to Arabic numbers

a) XXIV

b) LIX

c) XCIX

d) CCIX

e) XIX

6) Convert the following to Roman numerals

a) 179

b) 14

c) 77

d) 86

e) 111

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7) Which number between 1 and 100 would be the longest Roman numeral?

8) Which number would be the first that requires four different characters in Roman numerals?

9) Write a Roman numeral that contains more than one different character and is a palindrome

10) Which of the following Roman numerals is incorrect? Give the correct Roman numeral.

a) 40 = XXXX

b) 99 = IC

c) 95 = VC

d) 19 = IXX

e) 49 = XLIX

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Exercise 2

Place Value

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1) Write the following in numerals

a) Three hundred and twenty seven

b) Four thousand two hundred and twelve

c) Seven hundred and seven

d) Six thousand and fifteen

e) Twelve thousand four hundred and twenty

f) Thirty two thousand and eleven

2) Write the following in words

a) 3233

b) 41002

c) 706

d) 5007

e) 30207

f) 100001

3) What is the place value of the 5 in each of the following?

a) 1005

b) 51443

c) 75111

d) 523123

e) 54

f) 65121

4) Write the following numbers in order, from largest to smallest

121234, 11246, 13652, 834, 999, 1011, 1101,

5) Write the following numbers in order, from smallest to largest

4224, 425, 501, 5001, 516, 111, 1111, 11002, 1009

6) There were 26244 people at a soccer match. Write this number to the nearest

a) Hundred

b) Thousand

c) Ten thousand

7) Round the number 67532556 to the nearest:

a) Ten

b) Hundred

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c) Thousand

d) Ten thousand

e) Hundred thousand

f) Million

8) Add the following

a) 327 + five hundred and seventy five

b) Two thousand and nine + 747

c) Twenty thousand one hundred + eighteen thousand two hundred and twelve

d) 1143 + three thousand one hundred and two

e) 17111 + three hundred and ninety nine

9) Which numeral represents hundreds in the number 323468

10) If 50,000 is added to the number 486,400, which numerals change place value?

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Exercise 3

Factors & Multiples

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1) List the factors of the following numbers

a) 7

b) 9

c) 10

d) 12

e) 25

f) 30

2) By using a factor tree find the prime factors of the following

a) 16

b) 20

c) 64

d) 100

e) 144

f) 261

3) Find the greatest common factor of the following pairs of numbers

a) 2 and 6

b) 6 and 15

c) 10 and 25

d) 14 and 49

e) 12 and 64

f) 36 and 99

4) List all the multiples of the following that are less than 50

a) 3

b) 4

c) 5

d) 7

e) 10

f) 15

5) List the multiples of the following that are greater than 50 and less than 75

a) 2

b) 5

c) 6

d) 8

e) 11

f) 40

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6) Find the least common multiple of the following pairs of numbers

a) 2 and 3

b) 3 and 5

c) 4 and 6

d) 5 and 20

e) 6 and 32

f) 10 and 12

7) Jim writes the letter X on every 8th page of a book, while Tony writes the letter A on every 10th page.

a) What is the first page that has an X and an A?

b) What are the first 3 pages that have an X and an A on them?

c) If the book has 300 pages what is the last page in the book that has an X and an A?

8) A stamp collector has 24 Australian stamps, 40 English stamps, and 64 American stamps. If each page of his album has the same number of stamps, how many stamps are on each page, and how many pages are in the album? Note the stamps of different countries cannot be on the same page.

9) A loaf of bread contains 24 slices and a packet of ham has 5 slices. What is the smallest number of loaves of bread and packets of ham that must be bought to make **sandwiches** so there is no bread or ham left over? How many sandwiches will be made?

10) A light flashes every 6 seconds, and a horn sounds every 9 seconds. In two minutes how many times will the light flash and the horn sound at the same time?

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Exercise 4

Operations on Whole Numbers

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1) Add the following

a) 54 + 26

b) 17 + 47

c) 21 + 45

d) 19 + 55

e) 33 + 62

f) 72 + 22

2) Subtract the following

a) $99 - 54$

b) $83 - 32$

c) $67 - 46$

d) $71 - 51$

e) $84 - 13$

f) $57 - 45$

3) Add the following

a) 93 + 68

b) 64 + 46

c) 73 + 51

d) 112 + 103

e) 146 + 119

f) 163 + 104

4) Subtract the following

a) $274 - 162$

b) $312 - 153$

c) $422 - 113$

d) $812 - 333$

e) $713 - 618$

f) $901 - 565$

5) Multiply the following

a) 42×5

b) 33×8

c) 7×52

d) 11×13

e) 27×12

f) 31×15

6) Multiply the following

a) 34×27

b) 52×28

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c) 61×22

d) 53×41

e) 66×37

f) 71×19

7) Divide the following

a) $99 \div 9$

b) $84 \div 7$

c) $54 \div 6$

d) $78 \div 12$

e) $95 \div 4$

f) $86 \div 8$

8) Divide the following

a) $150 \div 15$

b) $220 \div 10$

c) $180 \div 20$

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Exercise 5

Unit Fractions: Comparison & Equivalence

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1) Which is the bigger fraction?

a) $\frac{1}{2}$ *or* $\frac{1}{5}$

b) $\frac{1}{7}$ *or* $\frac{1}{4}$

c) $\frac{1}{5}$ *or* $\frac{1}{6}$

d) $\frac{1}{3}$ *or* $\frac{1}{8}$

e) $\frac{1}{20}$ *or* $\frac{1}{10}$

2) Put the following in order from largest to smallest

a) $\frac{1}{5}, \frac{1}{2}, \frac{1}{3}$

b) $\frac{1}{7}, \frac{1}{3}, \frac{1}{6}$

c) $\frac{1}{9}, \frac{1}{10}, \frac{1}{2}$

d) $\frac{1}{2}, \frac{1}{11}, \frac{1}{5}$

3) John eats one-third of a cake and Peter eats one-fifth. Who has more cake left?

4) Debbie and Anne drive the same type of car and both go to the same petrol station at the same time to fill their petrol tanks. Debbie needs half a tank of petrol tank to be full, while Anne needs a quarter of a tank to fill up. Who will have to pay more for petrol

5) Bill and Ben start running at the same time. After one minute Bill has run one-quarter of a lap and Ben one-fifth of a lap. If they continue to run at the same speed, who will finish the lap first?

6) Which of the following fractions is the fraction $\frac{1}{2}$ equal to?

$\frac{3}{5}, \frac{3}{6}, \frac{3}{7}, \frac{2}{4}, \frac{4}{10}$

7) Four friends decide to share a pizza. If they each have an equal sized piece and eat all the pizza between them, what fraction of the pizza does each person get?

8) In a mathematics test Tom got $\frac{1}{4}$ of the questions wrong, and Alan got $\frac{1}{3}$ of the questions wrong. Who did better on the test?

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9) Josh and Tim are each reading a book. Josh's book has 10 chapters of which he has read 5, while Tim has read 4 out of 8 chapters. Who has read the greater fraction of their book?

10) Put the following fractions in order from smallest to largest

$\frac{1}{3}, \frac{2}{4}, \frac{1}{4}, \frac{1}{2}, \frac{3}{6}, \frac{1}{9},$

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Exercise 6

Operations on Decimals: Money problems

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1) Order the following from smallest to largest

0.4, 0.25, 0.33, 0.11, 0.05, 0.9, 0.09, 0.5, 0.01, 0.1

2) Order the following from largest to smallest

0.91, 0.19, 1.34, 0.34, 0.09, 1.91, 0.03, 0.05, 0.55, 1.55, 0.195

3) Add the following

a) 0.23 + 0.42

b) 0.15 + 0.62

c) 0.33 + 0.45

d) 0.71 + 0.28

e) 0.55 + 0.45

f) 0.8 + 0.3

4) Add the following

a) 0.58 + 0.36

b) 0.75 + 0.18

c) 0.22 + 0.69

d) 0.54 + 0.87

e) 0.99 + 0.51

f) 0.86 + 0.48

5) Add the following

a) 1.42 + 2.11

b) 1.61 + 0.22

c) 2.35 + 1.21

d) 4.23 + 1.62

e) 5.11 + 3.11

f) 1.55 + 1.56

6) Add The following

a) 2.67 + 4.44

b) 3.68 + 3.54

c) 2.59 + 4.62

d) 1.99 + 3.98

e) 6.77 + 3.25

f) 3.49 + 4.88

7) Subtract the following

a) 0.54 – 0.23

b) 0.86 – 0.13

c) 0.99 – 0.48

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d) 0.77 – 0.66

e) 0.12 – 0.02

f) 0.25 – 0.24

8) Subtract the following

a) 1.41 – 0.61

b) 1.89 – 0.92

c) 2.12 – 0.43

d) 3.24 – 2.56

e) 9.57 – 7.94

f) 2.15 – 0.99

9) Tom has $2.67 and lends Alan $1.41. How much money has Tom now got?

10) Francis buys a pen for $1.12, a ruler for $0.46 and a book for $5.20. How much did he spend in total?

11) At a fast food place, burgers are $4.25, fries are $1.60, drinks are $1.85, and ice creams are $0.55 each. How much money is spent on each of the following?

a) A burger and fries

b) A burger, drink and ice cream

c) Two burgers

d) Two fries and a drink

e) Two drinks and two ice creams

12) Martin gets $10 pocket money. He spends $1.65 on a magazine, $1.15 on a chocolate bar, $3.75 on food for his pet fish, and $1.99 on a hat. How much pocket money does he have left?

13) How much change from $20-should a man get who buys two pairs of socks at $2.50 each and a tie for $6.90?

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14) Peter needs $1.25 for bus fare home. If he has $5 and buys 3 bags of chips that cost $1.40 each, how much money does he have to borrow from his friend so he can ride the bus home?

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Year 5 Mathematics

Chance & Data

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Useful formulae and hints

The chance of an event happening range from 0 (impossible) to 1 (certain). A chance of ½ represents an event where there are two possible outcomes and each is as likely to occur as the other (Tossing a coin)

Graphs can show

- Changes over time
- Records of certain events (for example number of students getting 60% on a test)
- Quantities at a point in time

Different types of graphs are more suitable than others depending on the information to be shown

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Exercise 1

Simple & Everyday Events

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1) Put the following events in order from least likely to happen to most likely to happen

a) You will go outside of your house tomorrow

b) You will find a $100 note on the ground

c) The sun will rise tomorrow

d) You will pass a maths test you didn't study for

e) You will be elected President of the United States within the next year

f) You will toss a coin and it will land on heads

2) A boy's draw has 3 white, 5 black and 2 red t-shirts in it. If he reaches in without looking:

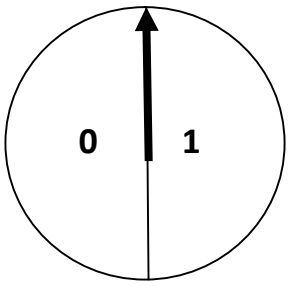
a) What colour t-shirt does he have the most chance of pulling out?

b) What colour t-shirt does he have least chance of pulling out?

c) What chance does he have of pulling out a blue t-shirt?

3) A man throws a coin 99 times into the air and it lands on the ground on heads every time. Assuming the coin is fair, does he more chance of throwing a head or a tail on his next throw? Explain your answer

4) A person spins the spinner shown in the diagram. If he does this twice and adds the two numbers spun together what total is he most likely to get?



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5) A man has 2 blue socks and 2 white socks in a draw. If he pulls out a blue sock first, is he more likely or less likely to get a pair if he chooses another sock with his eyes closed?

6) There are 10 blue, 10 green and 10 red smarties in a box. If a person takes one from the box without looking, which colour is he most likely to pull out? If he keeps pulling smarties out, how many smarties must he pull out in total to make sure he gets a green one

7) John thinks of a number between 1 and 10, while Alan thinks of a number between 1 and 20. Whose number do I have a better chance of guessing?

8) A set of triplets is starting at your school tomorrow. You do not know how many of them are boys and how many are girls. List all the possible combinations they might be.

9) Our school canteen has mini pizzas with three toppings on each one. The toppings are selected from:

- Ham
- Pineapple
- Anchovies
- Olives

a) What are the possible combinations of pizza available?

b) If I do not like anchovies, how many pizzas from part a will I like?

c) If EVERY pizza MUST HAVE ham as one of the three toppings, how does this change the answers to questions a and b?

10) On my lotto ticket I mark the numbers

1, 2, 3, 4, 5, 6

My friend's numbers are

12, 18, 19, 23, 27, 42

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Which one of us is more likely to win Lotto? Explain your answer

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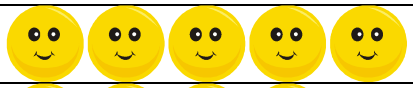
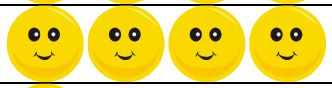
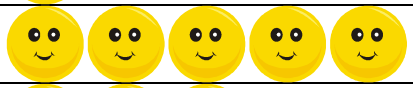
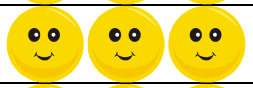
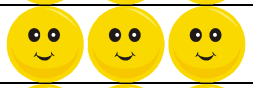
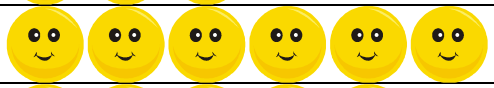
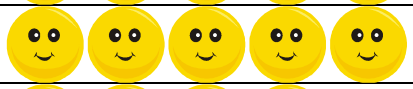
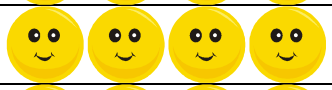
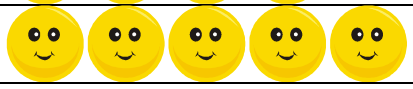
Exercise 2

Picture Graphs

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1) The picture graph below shows the approximate attendance at a soccer match for the past ten games

Each "face" represents 1000 people

Game Number	Attendance
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

a) For which game was there the largest crowd and what was the approximate attendance?

b) Which two consecutive games had approximately the same size crowd?

c) What was the most common attendance figure?

d) For one game the weather was cold and windy and there was a transport strike. Which game number was this most likely to be? Approximately how many people attended this game?

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2) The picture graph below shows the approximate number of fish caught at a beach over the past ten years. Each "fish" represents 500 fish

Year	Fish caught
2002	
2003	
2004	
2005	
2006	
2007	
2008	
2009	
2010	
2011	

a) Approximately how many fish were caught in 2003?

b) In which year were the most fish caught and how many was this?

c) In what year do you think the government put a restriction on the number of fish that could be caught?

d) How many fish have been caught in total over the past ten years?

3) The approximate average temperature for selected months for a city is shown in the picture graph below. Each  represents 10 degrees, each  represents 5 degrees

Month	Average daytime temperature
February	
April	
June	

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August	
October	
December	

a) Which are the hottest months of those shown?

b) Which are the coldest months of those shown?

c) What is the average temperature in October?

d) From this graph estimate the average temperature for this city in November

e) From the graph, is this city in the northern or southern hemisphere? Explain your answer

4) Jenny wanted to use a picture graph to show the number of people living in the 20 biggest cities in the world. Why would the following be a poor choice for a symbol?

 = 1 person

Propose a better choice

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5) A class took a survey of each student's favourite fruit and drew the following graph from their results. One piece of fruit equals one vote



a) What is the most popular fruit in this class?

b) How many students' favourite fruit is watermelon?

c) How many students are in the class?

d) The voting was from a list given to the students by their teacher. Nobody voted for a lemon as their favourite fruit. Discuss how this shows limitations of using picture graphs

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6) Draw a picture graph that shows the number of days it rained in a series of weeks from the table of data. Make up your own symbol and scale

WEEK NUMBER	NUMBER OF RAINY DAYS
1	2
2	4
3	0
4	6
5	7
6	4
7	5
8	3
9	2
10	0

7) What do you think the following picture graph is showing? (Hint: It is not showing size)

MY FAMILY

GRANDAD	
GRANDMA	

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DAD	
MUM	
ME	
BROTHER	
BABY SISTER	
PET DOG	

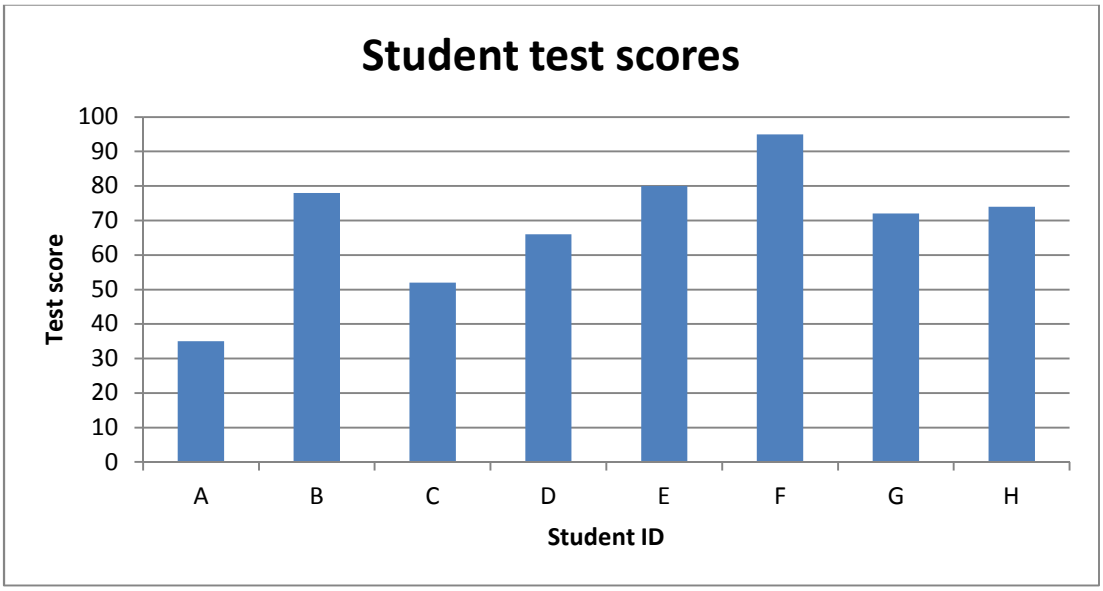
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Exercise 3

Column Graphs

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1) The following graph shows the test scores for a group of students



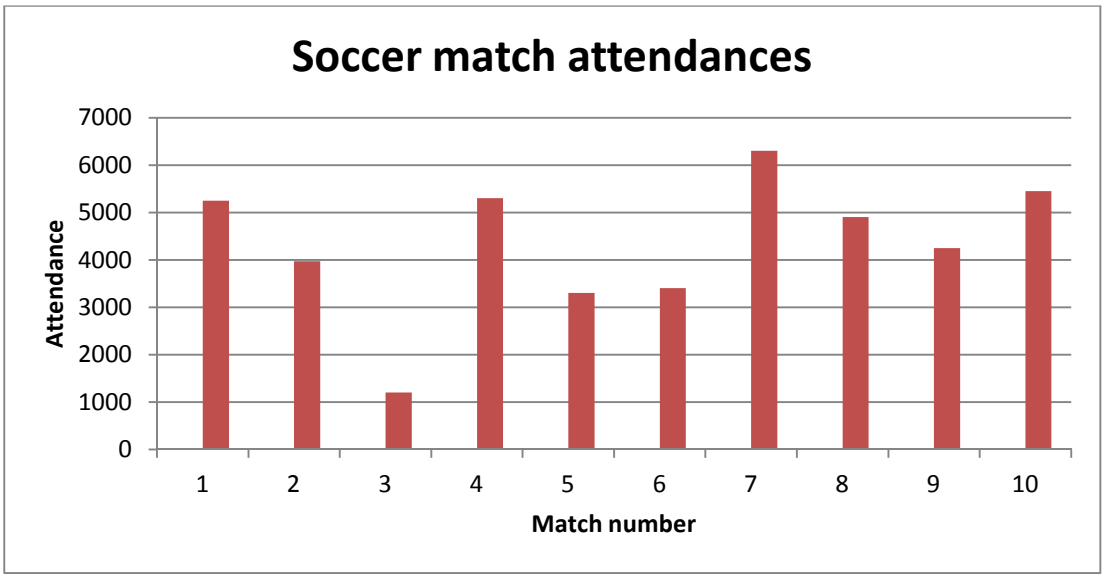
a) Which student scored the highest and what was their score?

b) How many students failed the test?

c) One student only just passed. What was their mark?

d) Name two students whose marks were almost the same

2) The attendances at the soccer matches from exercise 2, question 1 are shown in the column graph below



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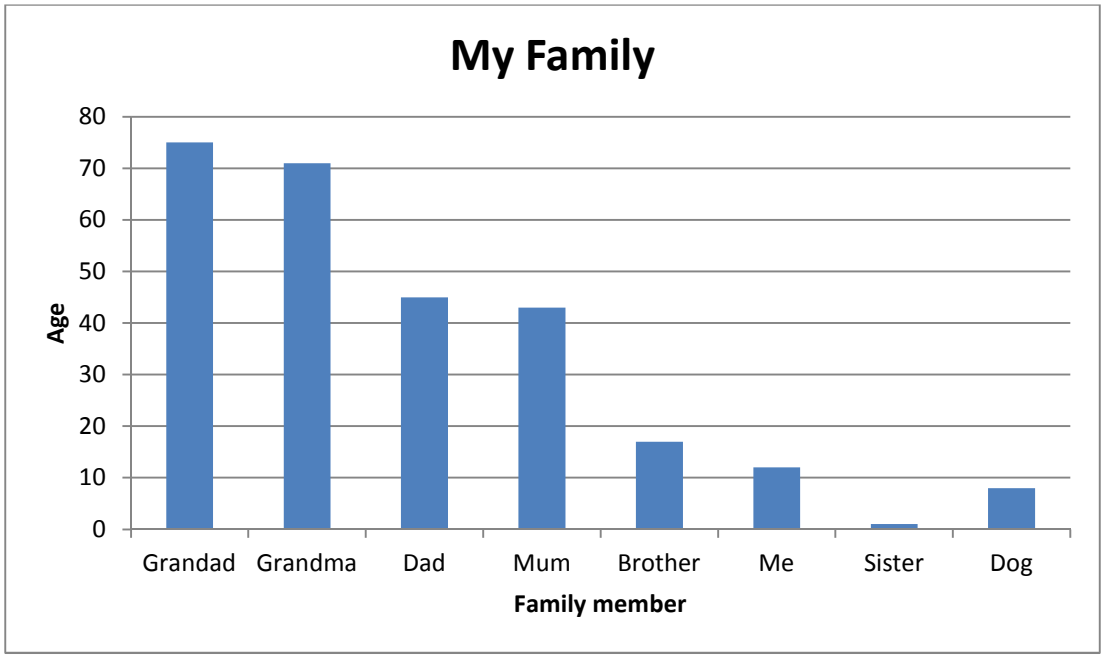
a) Estimate the attendance for game 1 and compare it with the estimate of the attendance using the picture graph from exercise 2

b) Repeat for game 10

c) What game had the highest attendance and approximately what was that attendance?

d) From your answers state an advantage of using column graphs over picture graphs

3) The following graph shows the ages of the members of a student's family



a) Who is the oldest in the family and how old are they?

b) Who is the youngest and how old are they?

c) Approximately how old is the dog?

d) How much older is the student's dad than the student?

e) From this question and the corresponding question in exercise2, discuss an advantage and a disadvantage of using column graphs to represent data

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4) Draw a column graph that represents the following data

Rainfall figures for week in mm

Day	Rainfall (mm)
Monday	22
Tuesday	17
Wednesday	9
Thursday	4
Friday	0
Saturday	11
Sunday	33

5) The following table shows the ten best test batting averages of all time (rounded to the nearest run)

Name	Average
Bradman	100
Pollock	61
Headley	61
Sutcliffe	61
Paynter	59
Barrington	59
Weekes	59
Hammond	58
Trott	57
Sobers	57

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Draw a column graph to represent the above data, and by comparing the data for Bradman to the others, discuss one advantage and one disadvantage of using column graphs to represent such a data set

6) The teacher of a large year group wishes to plot the ages of her students on a graph. Their names and ages are shown in the table below

Name	Average
Alan	12
Bill	12
Charlie	13
Donna	12
Eli	13
Farouk	12
Graham	12
Haider	13
Ian	13
Jane	13
Kate	12
Louise	12
Malcolm	13
Nehru	13
Ong	12
Paula	12
Quentin	13
Raphael	12
Sue	13
Tariq	13
Usain	13

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Veronica	12
Wahid	13
Yolanda	13

a) Plot the data on a column graph.

b) Imagine we had to graph the ages of year 7 students in the whole state. Using your graph as a guide, explain why a column graph is not suitable for displaying this data. Can you think of a better alternative?

7) A football club wanted to graphically show the ages of all players in their under 14 teams. Firstly they counted all the ages of the players and totalled the number of players of each age.

Age	**Number of players**
9	5
10	12
11	18
12	24
13	40

a) Draw this data as a column graph, and compare it to the column graph of question 6.

b) Which way of showing the players' ages graphically is easier to draw and shows the data in a smaller easier to read graph?

c) What is a disadvantage of graphing the ages in this way?

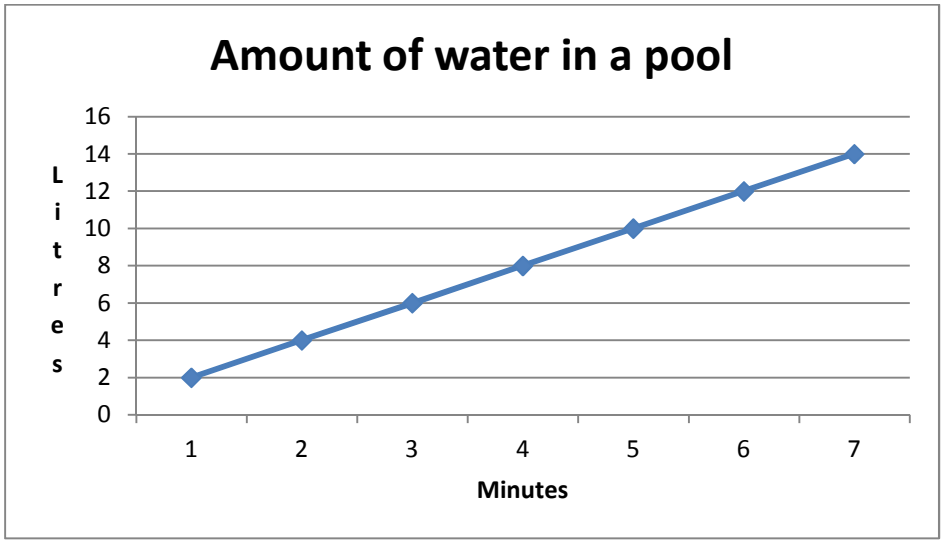
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Exercise 4

Line Graphs

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1) A pool is being filled with a hose. The graph below shows the number of litres in the pool after a certain number of minutes



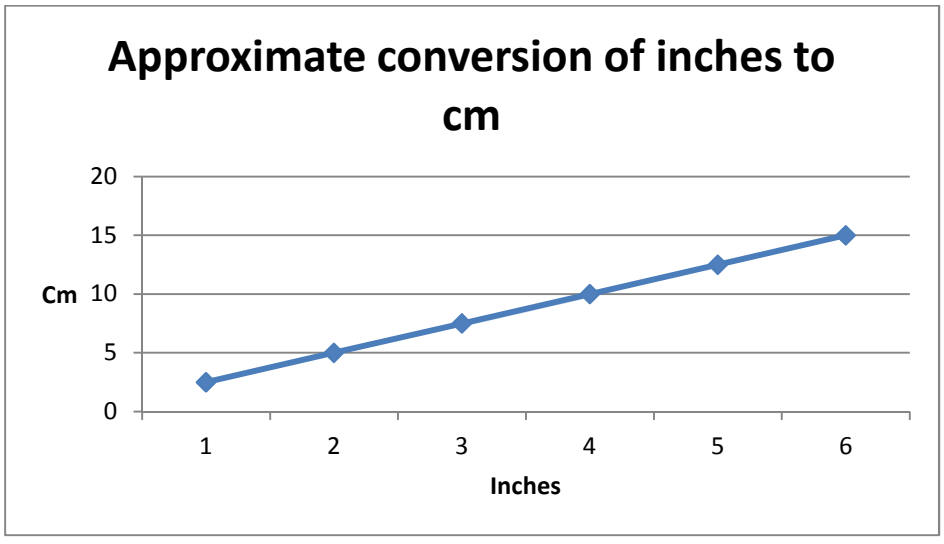
a) How much water was in the pool after 3 minutes?

b) How many minutes did it take to put 12 litres into the pool?

c) How fast is the pool filling up?

d) How many litres will be in the pool after 8 minutes, assuming it keeps getting filled at the same rate?

2) The graph below shows approximately how many cm are equal to a certain number of inches



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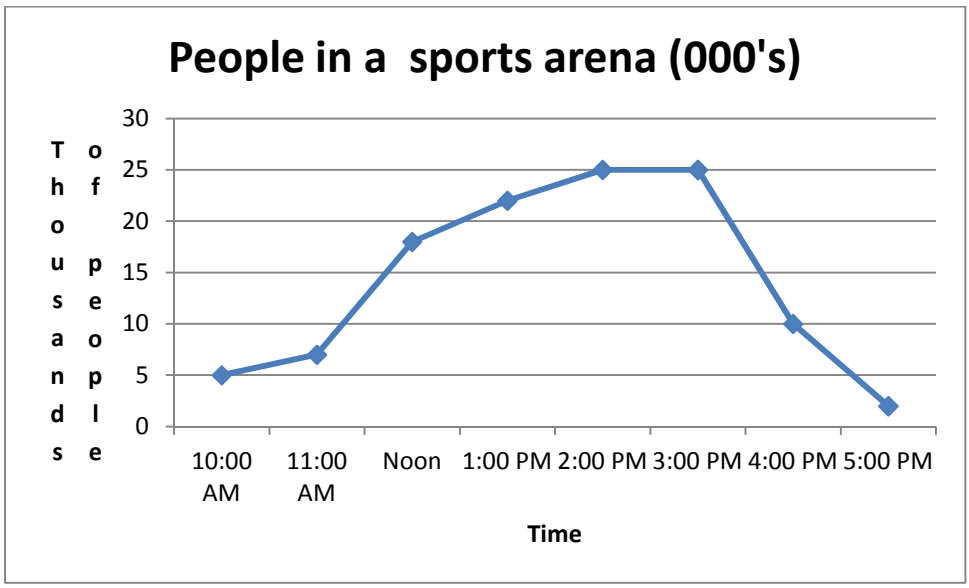
a) Approximately how many cm are there in 4 inches?

b) Approximately how many inches are there in 5 cm?

c) About how many cm equal one inch?

d) Approximately how many cm are in 8 inches?

3) The graph below shows how many people were at a sports arena at various times of the day



a) How many people were in the ground at 11 AM?

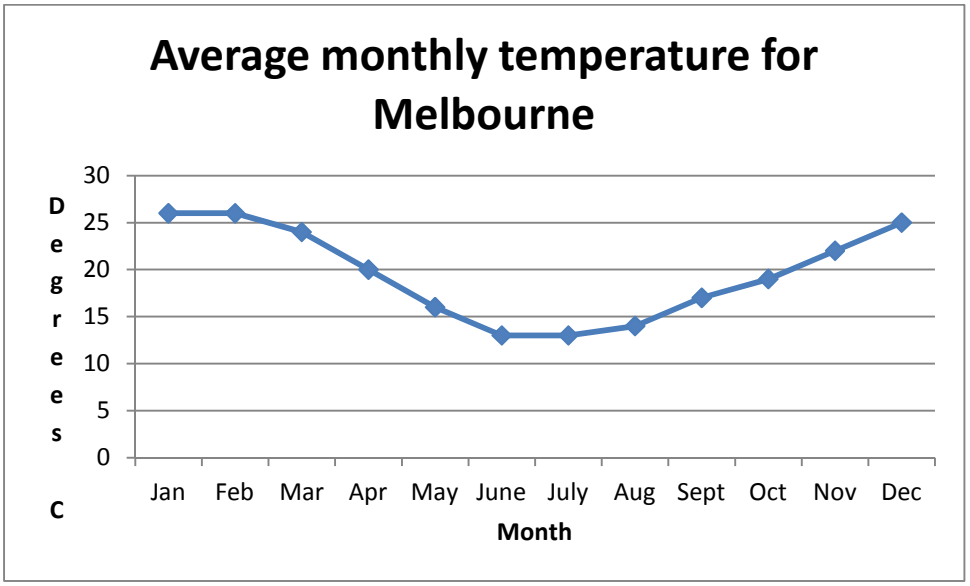
b) When were there approximately 10,000 people in the ground?

c) At what time would the game have started? Explain your answer

d) Why can't you say that the number of people in the ground at 3:30 PM was 15,000?

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4) The graph below shows the average daily temperature per month for Melbourne



a) What is the average daily temperature in December?

b) Which months are the coldest?

c) Name two non consecutive months when the average temperatures are the same

d) Does the graph show that temperatures in Melbourne will never go above 26 degrees? Explain your answer

5) Graph the following data in a line graph

Time	Number of people at a party
7 PM	6
8 PM	22
9 PM	30
10 PM	28
11 PM	25
Midnight	5

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6) Graph the following data in a line graph (Consider your scale)

Day	Number of buttons made at factory
Monday	6
Tuesday	8
Wednesday	11
Thursday	15
Friday	10
Saturday	5

7) Graph the following data that shows the population of Australia over time

Year	Population (approximate)
1858	1 million
1906	4 million
1939	7 million
1949	8 million
1958	10 million
1975	14 million
1989	17 million
2003	20 million
2008	22 million
2011	23 million

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Year 5 Mathematics

Algebra & Patterns

Useful formulae and hints

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Patterns represent changes in the relationship between two things. Called variables

Change can be

- Regular (Amount of water in a bath being filled at the same rate)
- Irregular (Change in population)
- Positive (Temperature of a heated pot)
- Negative (Amount of water in a bath after plug is pulled out)

Rules can be calculated and used to make predictions of future values

Rules can be calculated in two ways

1) How much one variable increases every time the other increases by the same amount

   For example: A pool starts off with 20 litres of water in it and is filled at the rate of 2 litres per minute. After one minute the pool has 22 litres, after 5 minutes the pool has 30 litres etc. A table is often useful in helping to determine these values.

2) A rule that relates one variable to the other, which is useful in predicting values where completing a table, would entail a lot of work. For example: in example one, to predict the amount of water in the pool after 200 minutes would require a large table and a lot of working out. If the rule that relates the amount of time to the amount of water in the pool can be worked out, the calculation is easier. The rule for the above is that the amount of water in the pool is equal to 20 plus two

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times the number of minutes it has been filling. Therefore after 200 minutes there would be 20 + (200 x2) = 420 litres

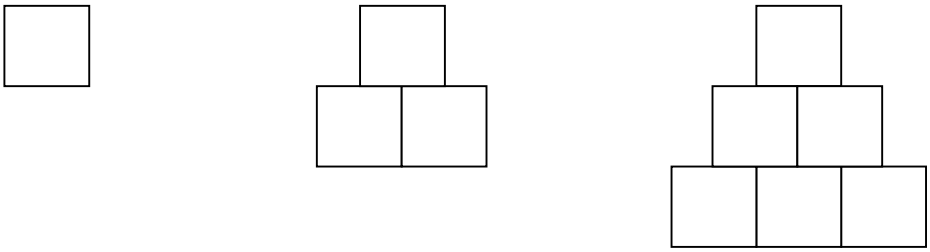
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Exercise 1

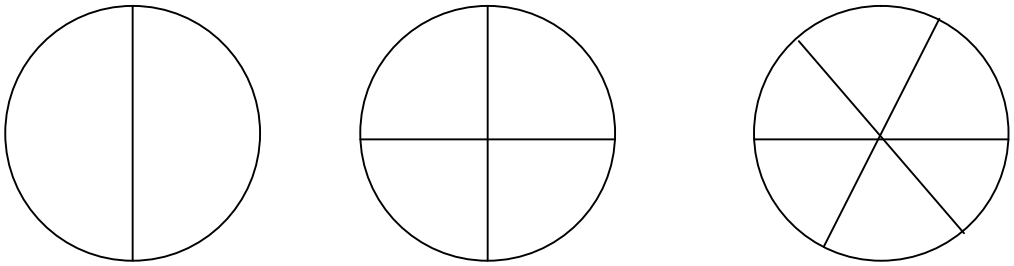
Simple Geometric Patterns

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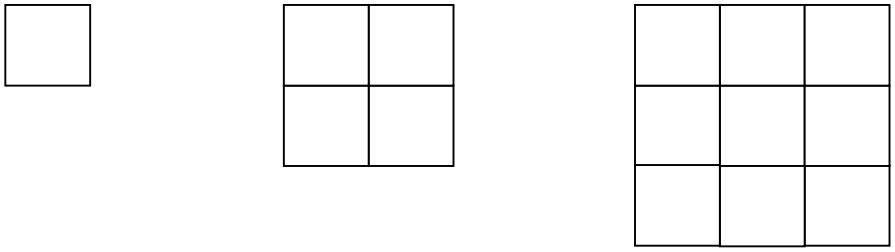
1) Draw the next two diagrams in this series



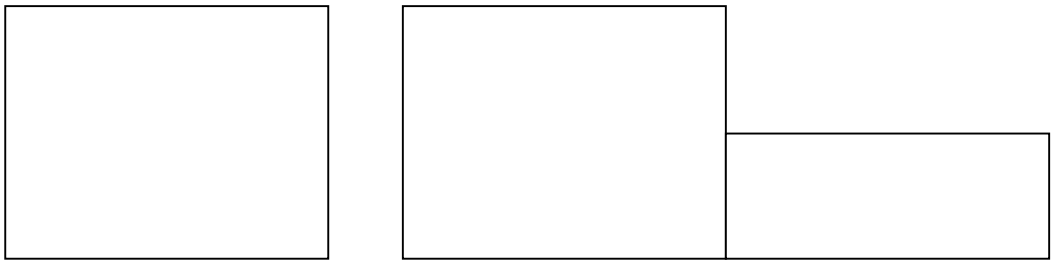
2) Draw the next two diagrams in this series



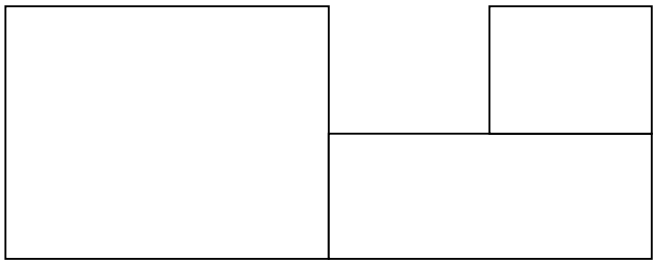
3) Draw the next two diagrams in this series



4) Draw the next two diagrams in this series

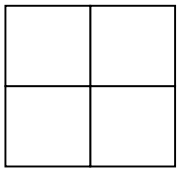


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5) To make two equal pieces of chocolate from a square block one cut is required. To make four equal pieces two cuts are required. How many cuts are needed to make 8 equal pieces? How many cuts are required to make 12 equal pieces?

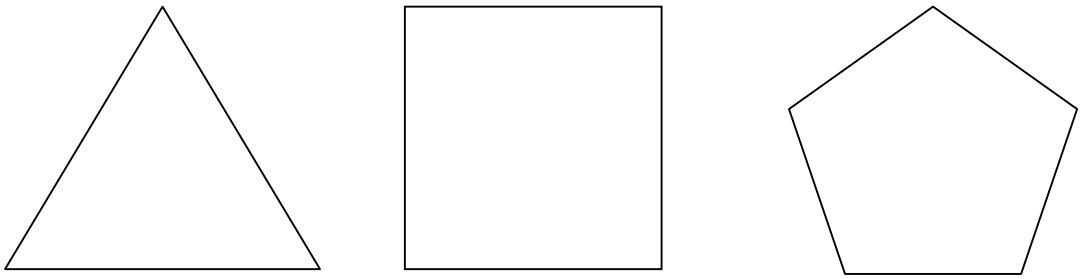
6) There are 5 squares on a 2 x 2 chessboard



Four small squares and one large square

How many squares on a 4 x 4 chessboard?

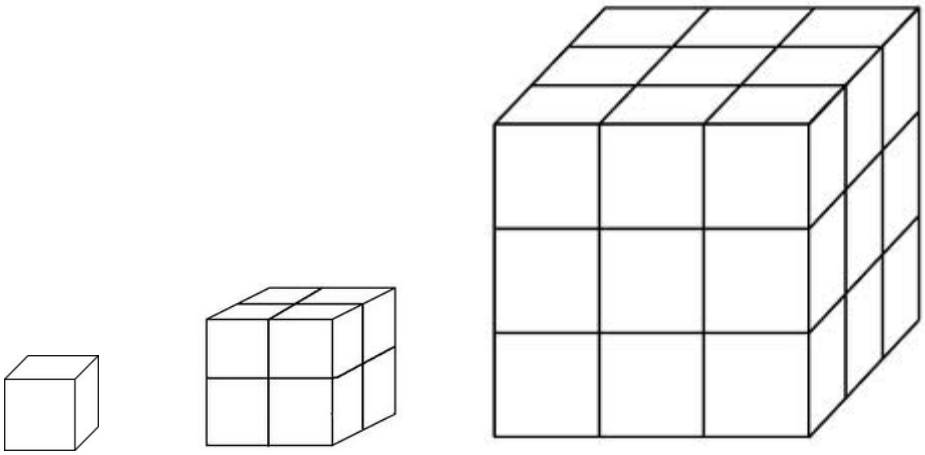
7) Measure and add up the internal angles of the following shapes



Use you results to predict the sum of the internal angles of a hexagon (6 sides) and a heptagon (7 sides)

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8) How many cubes in the next two shapes in this series?



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Exercise 2

Simple Number Patterns

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1) For the following series, fill in the next two terms

a) 1, 3, 5, 7

b) 2, 4, 8, 16

c) 1, 4, 9, 16

d) 1, 3, 6, 10

2) For the following series, fill in the next two terms

a) 5, 10, 15, 20

b) 32, 16, 8, 4

c) 100, 90, 80, 70

d) 64, 49, 36, 25

3) Fill in the blanks in the following

a) 2, 6, ___, 14, 18, ___

b) ___, 22, 33, ___, 55

c) 1, 3, ___, 27, ___, 243

d) 0.5, 1, 1.5, ___, ___

e) $\frac{1}{2}, \frac{1}{4}$, ___, $\frac{1}{16}$, ___

4) What are the next three numbers of the following series?

0, 1, 1, 2, 3, 5, 8

5) Thomas walked 3km on Monday, 6km on Tuesday, and 9km on Wednesday. If this pattern continues

a) How far will he walk on Friday?

b) What will be the total distance he has walked by Saturday?

6) At the start of his diet, a man weighs 110kg. Each week he loses 4kg.

a) How much weight will he have lost by the end of week 3?

b) How much will he weigh by the end of week 4?

7) A pond of water evaporates at such a rate that at the end of each day there is half as much water in it than there was at the start of the day. If there was 128 litres of water in the pond on day one, at the end of which day will there be only 8 litres of water left?

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8) Fill the blanks in the following series

a) 40, 42, 39, 43, 38, 44, ___, ____

b) 100, 200, 50, 100, 25, ___, ___

c) 1, ___, 10, 16, 23, ___

d) 1, 2, 5, 26, ___, ___

9) Complete the following series

a) 8, 12, 18, 27, ___

b) 4, 6, 10, 18, 34, ___, ___

c) 100, 60, 40, 30, ___, ___

d) 7.5, 7, 8.5, ___, 9.5, ___

10) A bug is crawling up a wall. He crawls 2 metres every hour, but slips back one metre at the end of each hour from tiredness.

a) How far up the wall will he be in 5 hours?

b) How long will it take him to reach the top of a 10 meter wall?

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Exercise 3

Rules of patterns & Predicting

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Different bacteria have different reproduction and death rates, so a group of different bacteria samples will have different populations depending on what type they are.

The populations of different types of bacteria were measured at one minute intervals, and the numbers present were recorded in separate tables which are shown in questions 1 to 7.

For each question you are required to:

- Fill in the missing figure
- Work out a rule that relates the number of minutes passed to the number of bacteria in the sample
- Use this rule to predict the number of bacteria in the sample after 100 minutes

The following example will help you

Minutes	1	2	3	4	10
Number	2	4	6	8	

It can be seen that the population increases by 2 bacteria every minute. Therefore in six minutes (the amount of time between 4 and 10), the population will increase by 12 bacteria (6 x 2). Therefore the population after 10 minutes will be 8 + 12 = 20 bacteria

To predict the population for longer time periods it is useful to find a rule that relates the number of minutes to the number of bacteria and apply that rule.

After 1 minute the population was 2 bacteria. This would suggest that if you add 1 to the number of minutes you will get the number of bacteria. The rule must work for every number of minutes. If you take 2 minutes and add 1 to it you get 3 bacteria, which does not match the table, therefore the rule is wrong

Another rule may be that you multiply the number of minutes by 2 to get the number of bacteria. This certainly works for 1 minute. What about 2 minutes or 3 minutes? If you multiply any of the minutes by 2 you will get the number of bacteria. Therefore you have found the rule.

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The rule should be stated:

The number of bacteria can be found by multiplying the number of minutes by 2

Use the rule to check your answer for 10 minutes found earlier (10 x 2 = 20, therefore correct), and to predict the number of bacteria after 100 minutes (100 x 2 =200)

NOTE: Some of the rules will involve a combination of multiplication and addition, or multiplication and subtraction

1)

Minutes	1	2	3	4	10
Number	4	5	6	7	

2)

Minutes	1	2	3	4	10
Number	3	5	7	9	

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3)

Minutes	1	2	3	4	10
Number	10	20	30	40	

4)

Minutes	1	2	3	4	10
Number	2	5	8	11	

5)

Minutes	1	2	3	4	10
Number	1	3	5	7	

6)

Minutes	1	2	3	4	10
Number	4	6	8	10	

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7)

Minutes	1	2	3	4	10
Number	110	120	130	140	

8) The time for roasting a piece of meat depends on the weight of the piece being cooked. The directions state that you should cook the meat for 30 minutes at 260 degrees, plus an extra 10 minutes at 200 degrees for every 500 grams of meat
How long would the following pieces of meat take to cook?

a) 500 grams of meat

b) 1 kg

c) 2 kg

d) 3.5 kg

9) Taxis charge a flat charge plus a certain number of cents per kilometre. A man took a taxi ride and noted the fare at certain distances

- After 1 km the fare was $2.50
- After 3 km the fare was $3.50
- After 10 km the fare was $7.00

What was the flat charge, and how much did each kilometre cost?

10) A business wanted to get two quotes to fix their truck, so they approached two different mechanics, Alan and Bob. Their quotes were:

Alan: $100 call out fee plus $40 per hour

Bob: $200 call out fee plus $20 per hour

Which mechanic should the company hire?

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Year 5 Mathematics

Measurement:
Length & Area

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Useful formulae and hints

There are 10 mm in one cm

There are 100 cm in one metre

There are 100 metres in one km

There are 100 square mm in one square cm

There are 10,000 square cm in one square metre

There are 10,000 square metres in one square km

The perimeter of a shape is the distance around its outside

The area of a rectangle or square is equal to length x width

The area of a triangle is equal to the length of the base x the perpendicular height, then halved

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Exercise 1

Units of Measurement

Converting & Applying

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1) Convert the following to metres

a) 3245 mm

b) 809 cm

c) 32 km

d) 5.43 km

e) 70 cm

2) Convert the following to centimetres

a) 41.4 m

b) 1762 mm

c) 4 m

d) 0.8 km

e) 9 mm

3) Convert the following to millimetres

a) 9 cm

b) 0.3 m

c) 1.27 m

d) 4 km

e) 19.2 m

4) Convert the following to square centimetres

a) 10 square metres

b) 100 square millimetres

c) 0.4 square kilometres

d) 0.142 square metres

e) 3174 square millimetres

5) Which is larger?

a) 145 mm or 1.45 cm

b) 73 km or 7300 m

c) 193 cm or 1930 mm

d) 10.3 m or 1030 mm

e) 0.5 km or 5000 cm

6) Which is smaller?

a) 144 square mm or 1.44 square cm

b) 1 square km or 100000 square metres

c) 178 square cm or 0.178 square metres

d) 100 square metres or 1000 square centimetres

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7) Each day for four days, Bill walks 2135 metres. Ben walks 1.2 km on each of five days. Who has walked the furthest?

8) Mark has to paint a floor that has an area of 180 square metres, whilst Tan has to paint a floor that has an area of 180000 square centimetres. Who will use more paint?

9) A snail travels 112 cm in 10 minutes, whilst a slug takes 20 minutes to go 22.4 metres. Which creature would cover more ground in an hour and by how much?

10) Alan walks 1.4 km to the end of a long road, then he walks another 825 metres to the next corner. He then walks 5 metres to the front of a shop and goes through the entrance which is 600 cm. How far has he walked altogether? Give your answer in km, m, and cm

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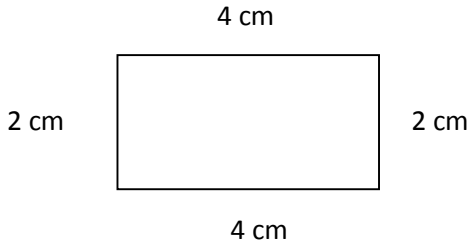
Exercise 2

Simple Perimeter Problems

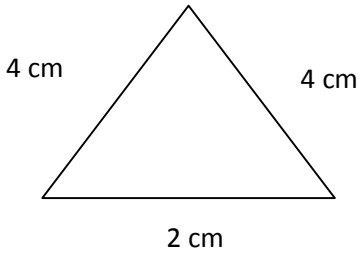
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1) Calculate the perimeter of the following

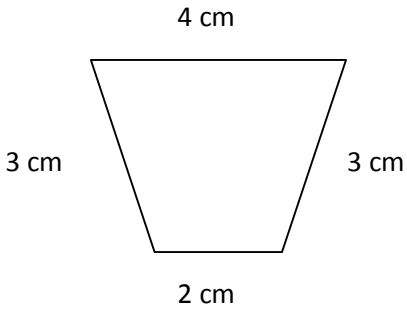
a)



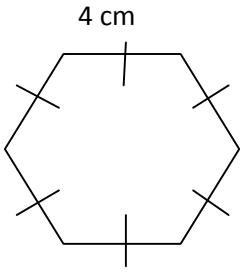
b)



c)

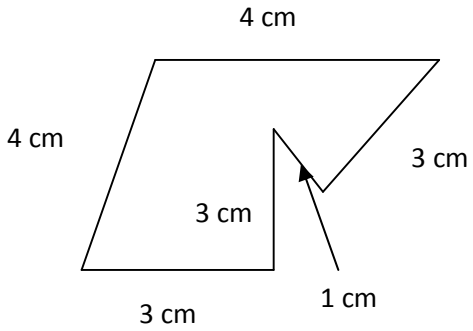


d)



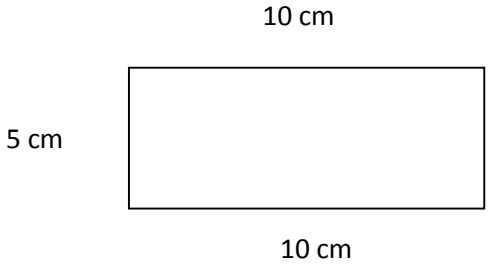
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e) A

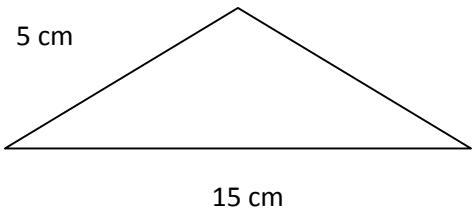


2) The perimeter of the following shapes is 30 cm. Calculate the unknown side length(s)

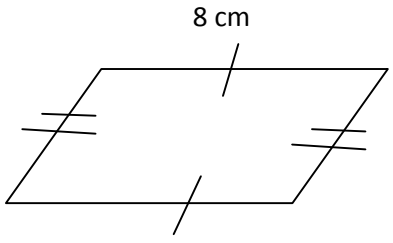
a)



b)

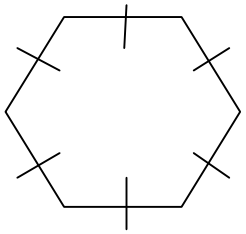


c)



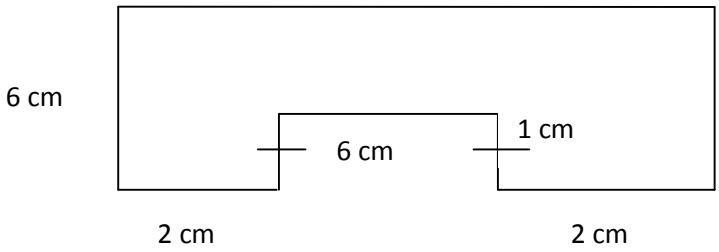
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d) A



3) A soccer field is 100 metres long and 30 metres wide. How far would you walk if you went twice around it?

4) Calculate the perimeter of the following shape



5) Two ants walk around a square. They start at the same corner at the same time. The first ant goes round the square twice while the second ant goes around once. In total they travelled 36 metres, what is the length of each side of the square?

6) What effect does doubling the length and width of a square have on its perimeter?

7) What effect does doubling the length of a rectangle while keeping the width the same have on its perimeter?

8) What must the side length of an equilateral triangle be so it has the same perimeter as a square of side length 12 cm?

9) The perimeter of a rectangle is 40 cm. If it is 6 cm wide, what is its length?

10) The length of a rectangle is 4 cm more than its width. If the perimeter of the rectangle is 16 cm, what are its measurements?

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11) Five pieces of string are placed together so they form a regular pentagon. Each piece of string is 8 cm long. How long should the pieces of string be to make a square having the same perimeter as the pentagon?

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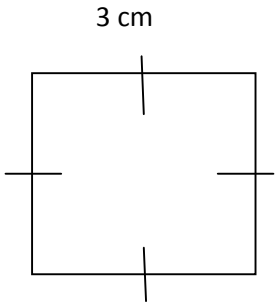
Exercise 3

Simple Area Problems

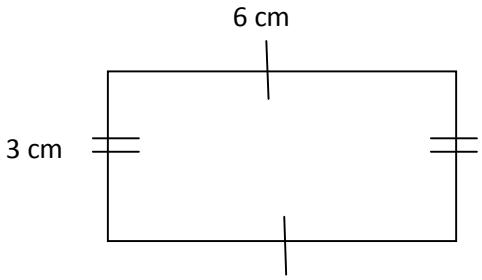
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1) Calculate the area of the following

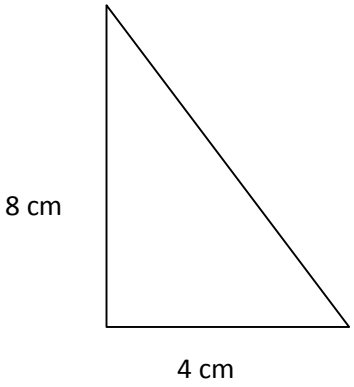
a)



b)

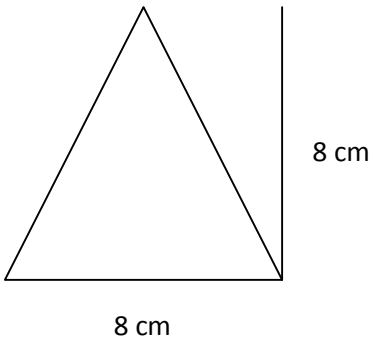


c)

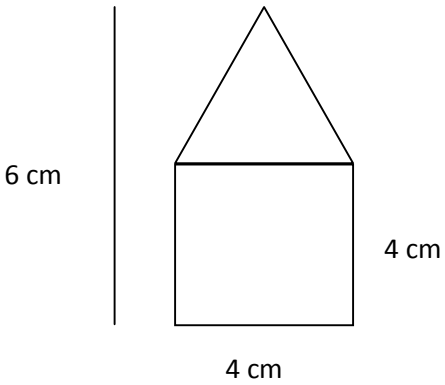


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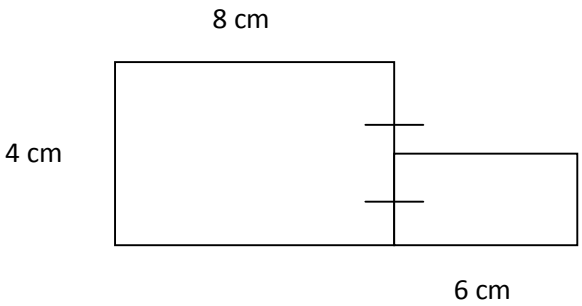
d)



e)



f)



2) A park measures 200 metres long by 50 metres wide. What is the area of the park?

3) The floor of a warehouse is 18 metres long and 10 metres wide. One can of floor paint covers 45 square metres. How many cans of paint are needed to paint the floor?

4) A tablecloth is 2 metres long and 500 cm wide. What is its area?

5) A wall measures 2.5 metres high by 6 metres wide. A window in the wall measures 1.5 metres by 3 metres. What area of the wall is left to paint?

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6) A customer requires 60 square metres of curtain fabric. If the width of a roll is 1.5 metres, what length of fabric does he require?

7) A square piece of wood has an area of 400 square centimetres. How long and how wide is it?

8) A stretch of road is 5 km long and 4 metres wide. What is its area?

9) A table is 400 centimetres long and 80 centimetres wide. What is its area in square metres?

10) A car park is 2.5 km long and 800 metres wide. What is its area in square metres and square kilometres?

11) Investigate the areas of rectangles that can be made using a piece of string that is 16 cm long. Complete the following table to help you. (Use whole numbers only for lengths of sides)

Length (cm)	Width (cm)	Area (cm^2)
1	7	7
2	6	12

12) A farmer has 400 metres of fencing in which to hold a horse. He wants to give the horse as much grazing area as possible, while using up all the fencing. Using your answers to question 11 as a guide, what should the length and width of his enclosure be, and what grazing area will the horse have?

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Year 5 Mathematics

Measurement:
Volume & Capacity

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Useful formulae and hints

There are 1000 cubic mm in one cubic cm

There are 1,000,000 cubic cm in one cubic metre

There are 1,000,000 cubic metres in one cubic km

One cubic cm equals 1mL

1000 mL equals one litre

One cubic metre equals one thousand litres

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Exercise 1

Determining Volume From Diagrams

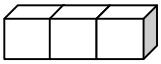
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1) Each cube in the following diagrams has a volume of $1cm^3$. Calculate the volume of the structure.

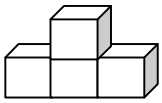
a)



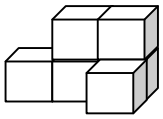
b)



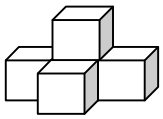
c)



d)



e)

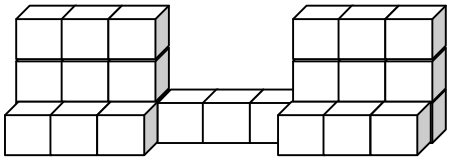


2) A wall is 5 blocks long, 3 blocks wide and 2 blocks high. Each block has a volume of $1m^3$. How many blocks are in the wall? What is the volume of the wall?

A diagram will assist you

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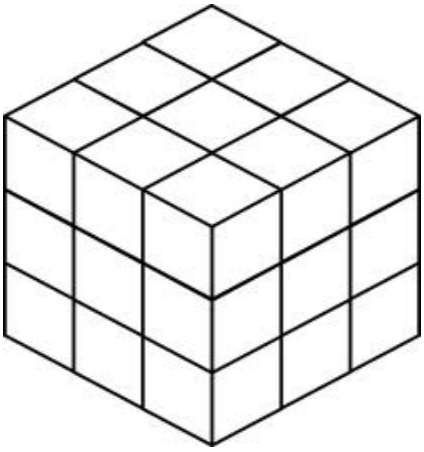
3) Each block in the following diagram has a volume of 0.5 cm^3, what is the volume of the structure?



4) The image below shows a chessboard; each square is a piece of wood that has a volume of 50 cm^3. Ignoring the border, what is the volume of the chessboard?



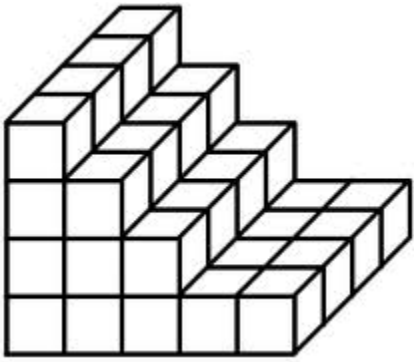
5) Each small cube that makes up the large one has a volume of 1 cm^3. What is the total volume of the large cube?



Use your result to show the general method of calculating the volume of a large cube.

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6) Each cube in the image below has a volume of 1 cm^3. What is the volume of the structure?



7) What is the volume of a stack of bricks each having a volume of 900 cm^3 if they are stacked 4 high, 5 deep, and 7 wide?

8) Three hundred identical cubes are made into a wall that is 3 blocks high, 5 blocks wide and 20 blocks long. If the total volume of the wall is 8,100,000 cm^3, what is the length of each side of one cube?

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Exercise 2

Units of Measurement: Converting & Applying

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1) Convert the following to cm^3

a) 1000 mm^3

b) 1 m^3

c) 2000 mm^3

d) 3500 mm^3

e) 0.1 m^3

2) Convert the following to m^3

a) 1,000,000 cm^3

b) 2,000,000 cm^3

c) 1 km^3

d) 0.1 km^3

e) 100,000 cm^3

3) A box has the measurements 100 mm x 100 mm x 10 mm. What is the volume of the box in cm^3?

4) A sand pit measures 400 cm x 400 cm x 20 cm. How many cubic metres of sand should be ordered to fill it?

5) Chickens are transported in crates that are stacked on top of and next to each other, and then loaded into a truck. Each crate has a volume of approximately 30000 cm^3.

How many crates could fit inside a truck of volume:

a) 300000 cm^3

b) 30 m^3

c) 270 m^3

6) A hectare is equal to 10,000 m^3. How many hectares in 1 km^3?

7) Put the following volumes in order from smallest to largest

10 m^3, 0.1 km^3, 5,000,000 cm^3, 10,000 mm^3

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8) Put the following in order from largest to smallest

100 cm^3, 10,000 mm^3, 0.01 m^3, 10 cm^3

9) A cube has a side length of 2000 mm. What is its volume in cm^3 and in m^3?

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Exercise 3

Relationship Between Volume & Capacity

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1) Convert the following to cm^3

a) 1 mL

b) 100 mL

c) 350 mL

d) 2 L

e) 10 L

f) 4.2 L

2) Convert the following to Litres

a) 1500 cm^3

b) 500 cm^3

c) 1250 cm^3

d) 10,000 cm^3

e) 100 cm^3

3) The following questions show the side length of a cube. Calculate the capacity of each cube in Litres

a) 10 cm

b) 100 cm

c) 500 cm

d) 1000 cm

4) The following questions show the capacity of a cube in Litres. What is the side length of the cube?

a) 1

b) 8

c) 27

d) 1000

5) Convert the following to Litres

a) 5 m^3

b) 10 m^3

c) 7.5 m^3

d) 3.52 m^3

e) 0.1 m^3

6) Convert the following to m^3

a) 500 L

b) 800 L

c) 3000 L

d) 10,000 L

e) 1550 L

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7) A swimming pool is 50 metres long by 10 metres wide, and has an average depth of 2 metres. What is the capacity of the pool in litres?

8) A swimming pool has a capacity of 500,000 litres. If it is 100 metres long by 5 metres wide, what is its average depth?

9) A water tank is 10 metres long by 8 metres wide by 10 metres deep. A chemical has to be added at the rate of one tablet per 200,000 litres. How many tablets need to be added to the tank?

10) Petrol sells for $1.50 per litre. A tanker carried $300,000 worth of petrol. The tanker was in the shape of a rectangular prism and measured 5 metres long and 4 metres deep. How long was the tanker?

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Year 5 Mathematics

Mass & Time

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Useful formulae and hints

There are 1000 mg in one gram

There are 1000 grams in one kilogram

There are 100 kilograms in one tonne

AM represents time between midnight and noon

Pm represents time between noon and midnight

The 24 hour clock shows the amount of time since midnight. For example, 1500 is 3 o'clock in the afternoon

When calculating elapsed time calculate the minutes elapsed first. If less than one hour, deduct one hour from the difference of hours

Example: Difference between 1:30 and 3:15

From 30 minutes to 15 minutes is 45 minutes

Is less than one hour, so deduct one hour from difference between 3 and 1. (3-1=2, 2-1=1)

Therefore time difference is 1 hour 45 minutes

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Exercise 1

Units of Mass Measurement:

Converting & Applying

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1) Convert the following to kilograms

a) 1000 g

b) 2000 g

c) 2500 g

d) 500 g

e) 750 g

f) 1.5 Tonne

g) 4 Tonne

2) Convert the following to grams

a) 1000 mg

b) 3000 mg

c) 2 kg

d) 3.5 kg

e) 600 mg

f) 100 mg

g) 100 kg

3) Convert the following to milligrams

a) 4 g

b) 10 g

c) 0.2 g

d) 1 kg

e) 100 g

4) A man places four 750 gram weights on one side of a scale. How many 1 kg weights must he place on the other side of the scale for it to balance?

5) Meat is advertised for $20 per kilogram. How much would 250 grams of the meat cost?

6) A rock collector collects 5 rocks. They weigh 300 grams, 400 grams, 500 grams, 1.5 kilograms, and 2 kilograms respectively. What was the total weight of his collection in grams and in kilograms?

7) A vitamin comes in tablets each of which has a mass of 200 milligrams. If there are 500 tablets in a bottle, and the bottle has a mass of 200 grams, what is the total weight of the bottle of tablets in grams and in kilograms?

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8) John has a parcel of mass 1.5 kilograms to send by courier. Courier company A charges $15 per kilogram, while courier company B charges 1.5 cents per gram. Which courier company is cheaper and by how much?

9) Which has more mass and by how much? Two hundred balls each with a mass of 100 grams, or 50 balls each with a mass of 0.5 kilograms.

10) A mixture has the following chemicals in it

- 1 kg of chemical A
- 750 g of chemical B
- 300 g of chemical C
- 800 mg of chemical D
- 700 mg of chemical E
- 500 mg of chemical F

What is the total mass of the mixture in kilograms, grams, and milligrams?

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Exercise 2

Estimating Mass

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1) For each of the following, state whether the usual unit of mass measurement is mg, g, kg, or tonnes

a) A human

b) Packet of lollies

c) An elephant

d) Loaf of bread

e) Paper clip

f) A car

g) An ant

2) A jack has a lifting capacity of 200 kg. Which of the following could be safely lifted by the jack?

- A truck
- A pool table
- A barbeque
- A spare tyre
- A carton of soft drink

3) Alfred buys a carton of butter that contains 10 x 375 gram tubs. What is the approximate mass of the carton to the nearest kilogram?

4) If a person rode on or in each of the following, for which would they increase the mass greatly?

- Horse
- Skateboard
- Bicycle
- Car
- Airplane
- Roller skates

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5) A car and a truck travelling the same speed each hit the same size barrier. Which one would push the barrier the furthest?

6) Put the following balls in order from smallest to heaviest mass

- Medicine ball
- Table tennis ball
- Tennis ball
- Golf ball
- Football
- Bowling ball

7) Approximately how many average mass adults could fit into a boat with a load limit of 1 tonne

8) Which has more mass; a kilogram of feathers or a kilogram of bricks? Explain your answer

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Exercise 3

Notations of Time

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1) Which of the following activities usually occur AM and which usually occur PM?

- Waking from a night's sleep
- Having dinner
- Going to school
- Having lunch
- Sport training
- Watching the sunset
- People working

2) School starts for Joseph at 9 AM and goes for 4 hours until lunchtime. At what time (AM or PM) does Joseph eat his lunch?

3) Write the time including AM or PM at one minute past midnight

4) Convert the following to AM or PM notation

a) 1030

b) 1115

c) 1515

d) 0200

e) 1600

f) 2120

g) 0725

h) 1925

5) Convert the following to 24 hour time notation

a) 3:00 PM

b) 1:15 AM

c) Midnight

d) 10:45 PM

e) 7:55 PM

f) Noon

6) Put the following times in order from earliest to latest

- 1515
- 3:10 AM
- 4:20 PM
- 1600
- 2020
- 11:22 AM

7) Charlie went to bed at 8:30 PM, Andrew went to bed at 1950, and Peter went to bed at 2040. Who went to bed earliest and who went to bed latest?

8) In Antarctica on the 7th December 2011, the sun rose at 0106 and set at 2351. Convert these times to

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AM and PM notation. What does your answer reveal to you?

9) Three people wrote down the following statements

- "I eat dinner at about 6 o'clock every evening"
- "I eat dinner at about 0715 every evening"
- "I eat dinner at about 1925 every evening"

Who was likely to have used the wrong time notation?

10) Three people are catching plane flights from the same airport on the same day. Andrew's flight leaves at 2:30 in the morning. Bob's flight leaves at 1510, and Chris' flight leaves at 2:58 PM. If check in is three hours before takeoff, who would have to arrive at the airport when their watch read AM time?

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Exercise 4

Elapsed Time, Time Zones

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1) How much time is there between the following pairs of times?

a) 1:15 AM and 7:20 AM

b) 4:35 PM and 8:50 PM

c) 9:12 PM and 11:59 PM

d) 4:25 AM and 6:40 PM

e) 11:44 AM and 6:51 PM

f) Noon and 3: 22 PM

2) How much time is there between the following pairs of times?

a) 0312 and 1133

b) 1533 and 1748

c) 1614 and 2217

d) 0830 and 1435

e) 1040 and 1853

f) 0958 and 1459

3) How much time is there between the following pairs of times?

a) 6:45 AM and 10:16 AM

b) 9:30 PM and 11:11 PM

c) 2:18 AM and 4:17 AM

d) 5:23 AM and 2:18 PM

e) 7:26 PM and 3:07 AM

f) 11:05 PM and 9:02 AM

4) How much time is there between the following pairs of times?

a) 0415 and 2:20 PM

b) 6:35 AM and 1543

c) 2120 and 2:25 AM

d) 0333 and 3:23 PM

e) 11:12 AM and 1601

f) 1117 and 3:07 AM the next day

5) A bus timetable states that bus number 235 leaves at 1525 and that the service runs every 35 minutes after that. What are the times of the next three buses (in 24 hour notation)?

6) Andre has to catch a train and a bus to get home. His train leaves at 1610, and arrives at the bus station at 5:05 PM. He waits ten minutes and catches the bus which takes 43 minutes to reach his stop. He then walks home for 5 minutes. How

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long does his journey take, and what time does he arrive home (Answer in both Pm and 24 hour notation)

7) The table below shows the time difference between some cities of the world.

City	Time difference (from Sydney)	Local time
Auckland	+ 2 hours	
Sydney	0 hours	0700
Hong Kong	-3 hours	
Paris		2100
London	-11 hours	
New York		1500
Los Angeles	-19 hours	

Complete the table

8) Perth summer time is three hours behind Sydney summer time. A plane leaves Sydney at 1400 Sydney time. The flight takes 4 and one half hours. What is the time in Perth when the flight lands?

9) From the table in question 7, if it is 4 PM on New Year's Eve in Los Angeles, what is the time and day in Sydney?

10) A man boards a flight in New York at 10 PM. The flight takes 7 hours to reach London. Using the table in question 7 as a guide, what time is it in London when the plane lands?

11) The circumference of the Earth at the equator is approximately 40070 km. Auckland and Paris are 12 hours apart in time. Using the knowledge that the Earth takes approximately one day (24 hours) to rotate once on its axis:

a) What is the approximate distance from Auckland to Paris?

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b) (Challenge Question): What is the approximate speed of the rotation of the Earth in kilometres per hour?

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Year 5 Mathematics

Space

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Useful formulae and hints

An equilateral triangle has all angles and all sides equal

An isosceles triangle has two sides equal as are the angles opposite them

A scalene triangle has no sides or angles equal

A right angled triangle has one angle of 90 degrees

A square has 4 sides all of which are equal in length, and which form right angles with each other

A rectangle has 4 sides, each opposite pair are equal in length, and parallel. The sides form right angles with each other

A rhombus has 4 sides; all of the same length; opposite sides are parallel. Opposite angles are congruent

A parallelogram has 4 sides; each opposite pair are equal in length and parallel. Opposite angles are congruent

A trapezoid has 4 sides, two of which are parallel

A prism is named after the shape that comprises its base and top; these are joined by rectangular sides

A pyramid has a triangular base

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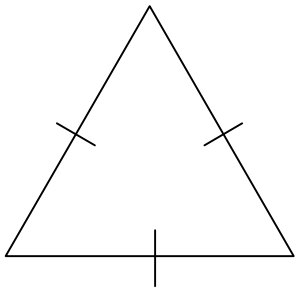
Exercise 1

Types & Properties of Triangles

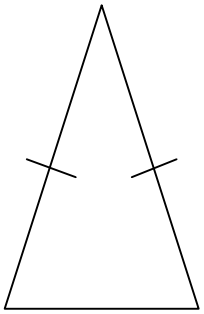
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1) Name the following triangles

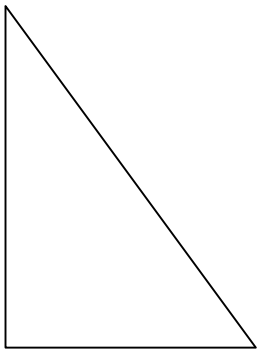
a)



b)

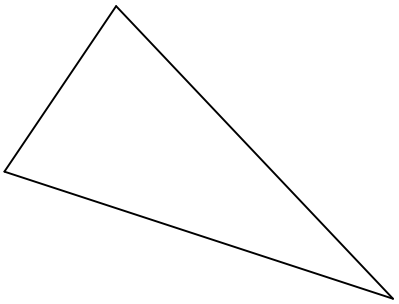


c)



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d)



2) True or false? The three angles of an isosceles triangle are congruent (the same size)

3) Which **types** of triangle can have two of its three sides equal?

4) Which type of triangle has two angles that are equal to 90 degrees?

5) Name two unique characteristics of an equilateral triangle

6) How many sides of an isosceles triangle are equal in length?

7) A triangle that has no sides equal in length is either a _____ triangle or a _____- triangle

8) If a square is cut across from one diagonal to another what type(s) of triangle(s) are formed?

9) If a rectangle is cut across from one diagonal to another what type(s) of triangle(s) are formed?

10) What is the size of each angle of an equilateral triangle?

11) If one of the angles of a right-angled triangle measures 60 degrees, what are the sizes of the other two angles?

12) Which type(s) of triangle(s) can have an angle greater than 90 degrees

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Exercise 2

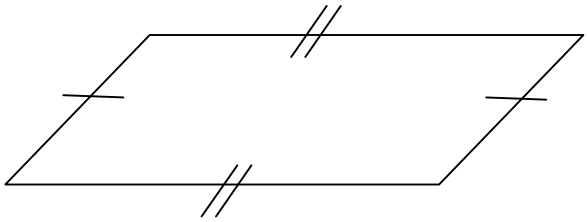
Types & Properties of Quadrilaterals

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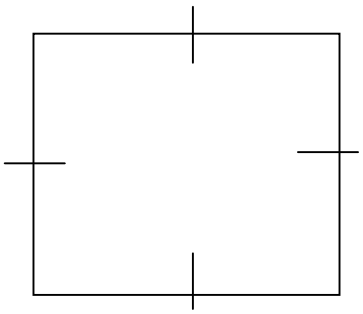
1) How many sides does a quadrilateral have?

2) Name the following types of quadrilaterals

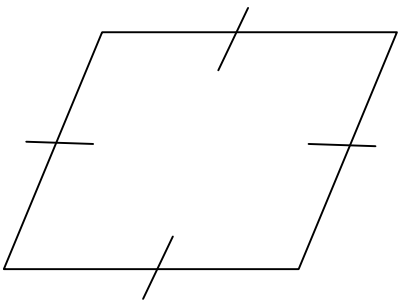
a)



b)

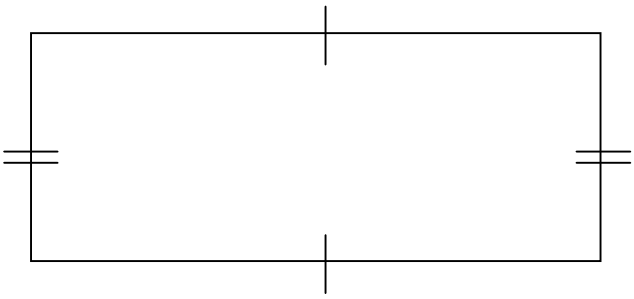


c)

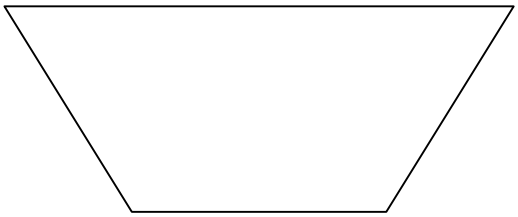


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d)



e)



3) Each angle of a square is _____ degrees

4) Name three quadrilaterals that have angles of more than 90 degrees

5) Name a quadrilateral that has a pair of sides not parallel

6) A rhombus is a special type of _____

7) A square is a special type of _____

8) Name three characteristics that are shared by a square and a rectangle

9) Name two characteristics that are shared by a trapezoid and a rectangle

10) Name the quadrilateral(s) that can have angles greater than 90 degrees

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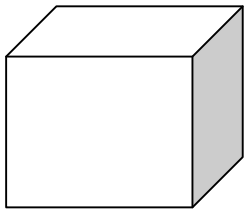
Exercise 3

Prisms & Pyramids

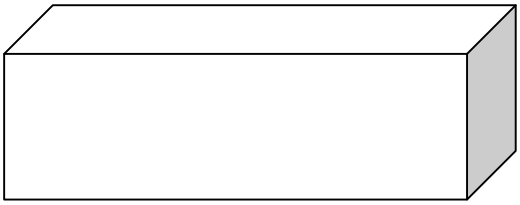
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1) Name each of the following shapes

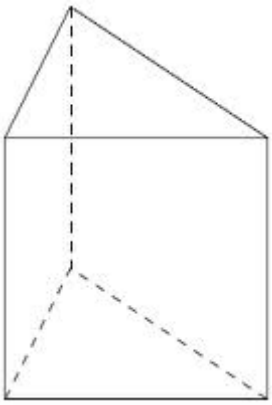
a)



b)

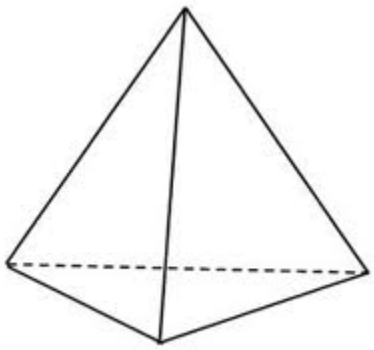


c)

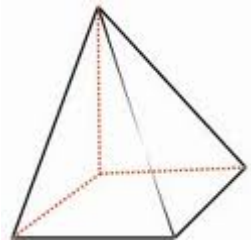


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d)



e)



2) What is the major difference between prisms and pyramids?

3) A shape has a hexagon at each end and rectangular sides joining them. What is this shape called

4)

a) How many faces does a rectangular prism have?

b) How many edges does a rectangular prism have?

c) How many vertices (corners) does a rectangular prism have?

5)

a) How many faces does a triangular pyramid have?

b) How many edges does a triangular pyramid have?

c) How many vertices does a triangular pyramid have?

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6)

a) How many faces does a triangular prism have?

b) How many edges does a triangular prism have?

c) How many vertices does a triangular prism have?

7) From your answers to questions 4 to 6, is there a rule that connects the number of faces, edges and vertices in a prism or pyramid?

8) All prisms have at least _____ pair of parallel faces

9) Pyramids have _____ pairs of parallel faces

10) What is the main feature of a cube that distinguishes it from other prisms?

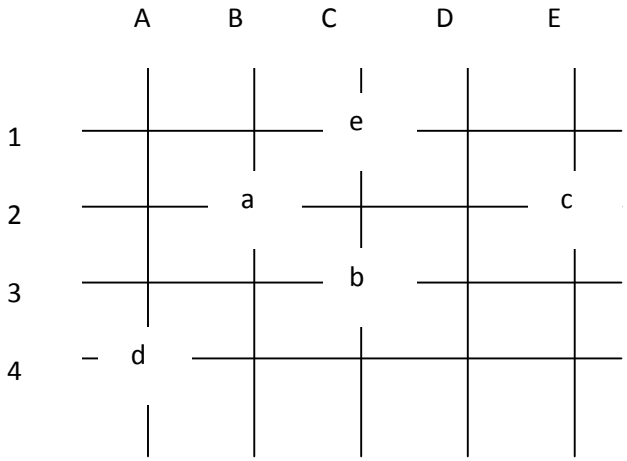
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Exercise 4

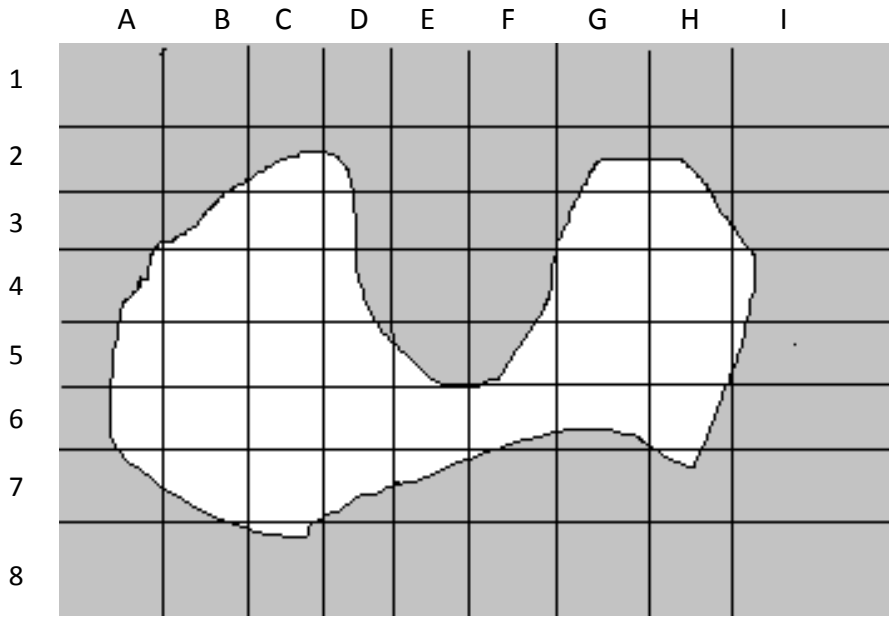
Maps: Co-ordinates, Scale & Routes

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1) Using the grid below, write the co-ordinates of the points a to e



2)



Mark the following co-ordinates on the map

a) D6

b) F7

c) C3

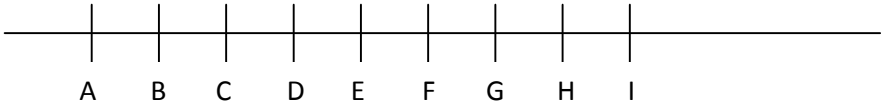
d) B5

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e) If the white portion of the map represents land and the grey represents water, give the co-ordinates of a square:

I. That is all land
II. That is all water
III. That is approximately half land and half water
IV. That is mostly land
V. That is mostly water

3)



The distance between each mark on the line represents 50 km. What distance is represented from:

a) A to D

b) B to E

c) B to G

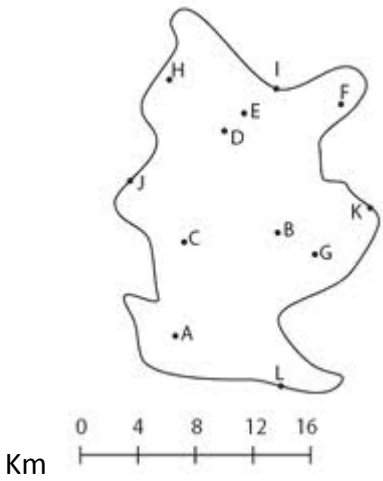
d) H to C

e) A to F and back to D

f) G to C and back to E

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4) Use the map and scale below it to answer the questions



What are the distances from:

a) Points A and H

b) Points C and K

c) Points F and D

d) Points B and G

e) Points L and K

5) The map below shows the Murray River and the south eastern portion of Australia



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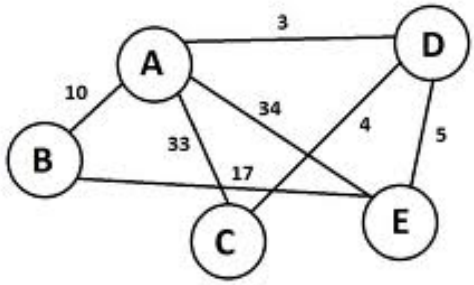
a) What is the approximate distance from Brisbane to Sydney?

b) What is the approximate distance from Canberra to Melbourne?

c) Approximately how long is the border between New South Wales and Queensland?

d) By treating the state of New South Wales as a rectangle, estimate its area.

6)



The diagram shows the shortest distance between any two points

a) Along which path or paths is the shortest distance from A to E?

b) What is the shortest distance from B to C?

c) What is the shortest distance from D to E if you must also go through point A?

d) What is the shortest distance if you must start at point A, visit each point once but only once and return to point A?

7) Draw a scale map that has the following information

a) A scale of 1 cm equals 10 km

b) The distance from A to B is 30 km

c) Point B is located at co-ordinate A5

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d) The distance from point A to point C is 50 km, but is 70 km if you go via point B

e) Point D is an equal distance (25 km) from points A and C

f) The points all lie on an island that is in the approximate shape of a rectangle and has an area of 2000 km^2

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